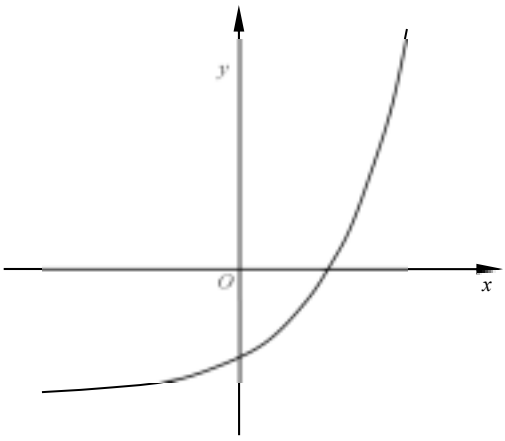
Question	Answer	Marks	Partial Marks
10(a)	–2.5 –2 –1	**3**	**B1** for each
10(b)	Correct curve 	**4**	**B3 FT** for 8 or 7 correct plots **B2 FT** for 6 or 5 correct plots **B1 FT** for 4 or 3 correct plots
10(c)	2.3 to 2.4	**1**	
10(d)	ruled line $y = x - 1.5$	**M2**	**M1** for $y = x - 1.5$ soi or for $2^x - 3 = x - 1.5$ seen. or $y = x + k$ or $y = kx - 1.5$ drawn Do not accept $y = -1.5$
	–1 and 1.55 to 1.7	**A2**	**A1** for each
11(a)	10	**3**	**M2** for $(1 - -7)^2 + (4 - -2)^2$ oe or **M1** for $(1 - -7)$ or $(4 - -2)$ oe
11(b)	$\frac{4}{3}$ or $\frac{8}{6}$	**2**	**M1** for $\frac{1 - -7}{4 - -2}$ oe
11(c)	$y = -\frac{3}{4}x - \frac{9}{4}$ or $4y + 3x + 9 = 0$ oe **final answers**	**4**	**B3** for $-\frac{3}{4}x - \frac{9}{4}$ OR **B1** for midpoint (1, − 3) **M1** for gradient $-\frac{3}{4}$ or $-\frac{1}{\textit{their}\ \textbf{(b)}}$ **M1** for substituting *their* (1, −3) into $y = (\textit{their } m)x + c$ or for $\textit{their } m = \frac{y - -3}{x - 1}$ oe
12(a)	$4x^3 - 16x$ cao	**2**	**M1** for $4x^3 + kx$ or $kx^3 - 16x$ or $4x^3 - 16x + k$ or $4x^3 - 16$ as final answers

Question	Answer	Marks	Partial Marks
12b	*Their* $\frac{dy}{dx} = 0$ or stating $\frac{dy}{dx} = 0$	**B1**	
	Correct method to solve *their* $4x^3 - 16x = 0$	**M1**	e.g. $4x(x^2 - 4)$ or $4x(x - 2)(x + 2)$ oe
	[x =] 0, –2, 2	**A1**	Or **B1** for (–2, –11) and (2, –11)
	(0, 5) (–2, –11) (2, –11)	**A1**	
12(c)	(0, 5) with correct reasoning	**2**	**M1** for any of • correct use of 2nd derivative $12x^2 - 16$ • evaluates correctly both values of y on either side • evaluates correctly the gradient on either side • reasonable correct sketch